

Running AI Locally – Daily Task Tutorial

Day 1 – Core AI Foundations

- Understand what AI really is (no hype, no math)
 - Learn the AI pipeline: Input → Thinking → Action → Output
 - Difference between LLM and SLM
 - Why small models are better for beginners
- Outcome: Clear mental model of AI systems

Day 2 – APIs & Backend Basics

- What an API is (waiter analogy)
- Why APIs exist (security, scalability, decoupling)
- HTTP methods (GET, POST, PUT, DELETE)
- API request–response lifecycle

Outcome: Understand how clients talk to servers

Day 3 – FastAPI Fundamentals

- What FastAPI is and why it is fast
 - ASGI vs WSGI
 - Creating virtual environments
 - Running FastAPI with Uvicorn
 - Auto-generated Swagger docs
- Outcome: Ability to build and run APIs

Day 4 – OpenAI API Integration

- What OpenAI API provides
- API key generation and security rules
- Environment variables (.env)
- Using Responses API (new standard)
- Text generation basics

Outcome: Cloud AI working safely

Day 5 – Wrapper API Pattern

- FastAPI as a middleman
 - Client → API → AI → API → Client
 - Why enterprises use wrapper APIs
 - Error handling and validation
- Outcome: Enterprise-safe AI architecture

Day 6 – Running AI Locally with Ollama

- What Ollama is
- Installing Ollama on Windows (CPU)
- Downloading and running llama3
- Understanding local AI servers
- No API keys required

Outcome: Offline, private AI system

Day 7 – Local AI APIs (Ollama API)

- Ollama REST API overview
- Chat endpoints
- Model management (pull, list, delete)
- Streaming responses

Outcome: Programmatic local AI usage

Day 8 – Tools & Safety

- AI thinks, tools act
 - File tools, image tools, voice tools
 - Why AI must not have full permissions
 - Treat AI like a smart intern
- Outcome: Secure AI system design

Day 9 – Memory & Vector Databases

- Why AI forgets
- Short-term vs long-term memory
- Prompt memory vs vector DB
- Document-based AI assistants

Outcome: Knowledge-aware AI

Day 10 – Final Architecture & Use Cases

- Full beginner-to-enterprise architecture
- Local ChatGPT clone
- Company document assistant
- Code helper
- Voice FAQ bot

Outcome: Production-ready AI mindset